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Beneficial Effects Coincident with Adding Olive Oil and Inositol in a Dog Diet: Cause and Effect, Chain of Events

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(Dated: April 11, 2026)

An earlier manuscript [69] described results of feeding a set of supplements to dogs with a particular focus on copper. One objective of that work was to return a dog with a heart condition, Happy, back to a high-energy state free from coughing as she briefly experienced in early 2019. This was not achieved so a closer look at the 2019 diet was performed. That effort pointed to other possible contributors that had been neglected for the last few years. These intially included extra virgin olive oil(EVOO), inositol, PABA, and silicon dioxide. Later, liquid sunflower lecithin that had been used earlier was also included. Creatine and carnitine were also introduced even though they had not been used before. Adding some of these components did appear to help. The initial change including olive oil improved Happy's energy and the feeding eagerness for 2 of the other dogs. There is some indication Happy was coughing less after about 6 weeks although after 3 months she continues to cough frequently. Addition of the other components and reduced use of the PABA did not produce notable changes. While the specific components have not been tested in isolation, the combination along with the background diet appears to work, probably due mostly to the oleic acid in the extra virgin olive oil and liquid lecithin. The literature on these supplements is suggestive of benefits but was not conclusive or compelling. It is likely that some anecdotal benefits can be realized with a better understanding of cause and effect and the role of the background diet. As of Feb 2026, all dogs are stable at near youthful levels although Happy continues to cough. Implications of these results may be relevant to resolving issues in topics ranging from dog nutrition to human diseases such as Alzheimer's.

Keywords: inositol; olive oil; sunflower; lecithin; PABA; canine nutrition ; creatine; alzheimer's

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CONTENTS

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I. INTRODUCTION

A recent manuscript detailed the overall safety and possible benefits of copper supplemetation in a group of dogs with heart benefits thought to be the most relevant [69]. However, results were not entirely successful particularly in the case of Happy, thought to have an enlarged heart, who never achieved ”cough free” status as she had briefly done in early 2019. Further lack of progress, with the hope that any disease progression was not entirely irreversible, motivated a more detailed examination of the last known working diet. Three differences were initially found between recent diets and the working 2019 diet that may have been important. First, the extra virgin olive oil (EVOO or simply olive oil herein) had been mostly eliminated except as an undisclosed liquid to distribute vitamin D supplements. Secondly, the earlier diet had used a multi-B vitamin known more or less as Big-B or B-100 or GNC-100 [3]. Most of the active ingredients from the label, pictured in Fig. 1, had been replaced by pure versions except for choline, inositol, and PABA. None of the ”Other” ingredients had been intentionally replaced although some appear in other vitamin formulations. Finally, the details of the sunflower lecithin had changed. Initially a capsule form was used before switching to a powder and intermittent use of a liquid form. No effort was made to revert to the capsule form but the liquid form was used again.

FIG. 1. Label from a recent version of the GNC multiB product similar to original. The mcg units on inositol is thought to be a mistake although no attempt was made to investigate.

FIG. 2. Label from the NOW Sunflower liquid lecithin which replaced their similar product of powdered lecithin.

This work describes the results of adding back these neglected components as well as a few others with possible benefits. When the diet had been in use for a little over a month with interruption, eating eagnerness had improved with overall energy level. Only three of the dogs were notably ”off” or ”starting to act old” but none in particularly bad shape. Dexter had initially been a picky eater but much better with zinc and now more consistent and eager. Rocky too would often prefer to sleep through breakfast now tends to eat even when cold in the morning although he also seems to be ”playing games” indicative of mental changes. He reverted back to later feedings but typically runs to the meal although may try to get me to move his feeding location. Happy may be getting some cough reduction with obviously more youthful energy level. 2 of the dogs, Annie and Trixie, have died since the last report and prior to significant consumption of these new components. Some current and future implications of these results are discussed as at a minimum they contribute more data points for specific dog diet results.

II. DISCOVERY AND CONSIDERATION OF SUSPECTS

An inability to entirely return Happy to her high-energy cough-free state motivated a more careful examination of the last known working diet from early 2019. The two most obvious suspects were extra virgin olive oil (EVOO) and ignored components of the GNC B-100 supplement. The latter containing many of those considered below , see the pictureFig. 1. Additional components known or suspected to be beneficial under some conditions such as creatine and carnitine were also added later in small amounts.

II.1. Extra Virgin Olive Oil

The prior olive oil dosing history is summarized in Table I and annual exposure in Table II. 2019 was clearly the year with the most consistent use and it tapered off a lot by 2024. Note of course that there are time lags between deprivation and symptoms and the dynamics of storage and usage are not known.

Days	End	Start	Comments
148	18833 2021-07-25	18685 2021-02-20	
75	18977 2021-12-16	18902 2021-10-02	
83	19225 2022-08-21	19142 2022-05-30	
98	19891 2024-06-17	19793 2024-03-11	
97	19762 2024-02-09	19665 2023-11-04	
65	19621 2023-09-21	19556 2023-07-11	
122	20029 2024-11-02	19907 2024-07-03	
282	20311 2025-08-11	20029 2024-11-02	
970	total days covered		

TABLE I. Notable gaps in olive oil consumption of any amount. Of the 2650 days during Happy's stay here she had olive oil only 589.

Year	Olive Oil Days
2018	93
2019	161
2020	62
2021	53
2022	63
2023	64
2024	22
2025	84

TABLE II. Number of days for each year of any olive oil for Happy. Note that 2024 was particularly low in olive oil and almost all of 2025 was in the last few months.

Olive oil is a complex natural product and a perennial suspect for healthy diets with many questions and paradoxes [34] [25]. A lot of motivation for study comes from its inclusion in the Mediterranean Diet and its complex composition [96].

Thinking outloud: the above claims to look at liquids in Med Diet but fails to mention vinegar, just EVOO and wine ...

Currently most health benefits are thought to relate to the monounsaturated oleic acid which is the majority of extra virgin olive oil although many other components are potentially active such as polyphenols and hydroxytyrosol [65] [71] . Specifics may vary among sources and be modified by storage or processing conditions [8] as with many natural products. While oleic acid is itself a fatty acid, it may paradoxically act to minimize pathological fat accumulation or facilitate lipid homeostasis.

Thinking outloud: like with metals that seem to facilitate distribution of each other

In a small lab study in rats, it was concluded that, " [...] the intake of dietary high OA [oleic acid] may enhance the omega-3 fatty acid levels in the blood plasma of rats and may have a positive effect in reducing risks to cardiovascular disease, as evidenced by weight loss, increased HDL-C levels, and decreased TG levels in the blood plasma of experimental animals" [75]. Lipid homeostasis and possible health and longevity improvements may be achieved through endoplasmic reticulum and nuclear mechanisms [19] . It may also act to improve skeletal muscle cell phenotype. Mice supplemented with oleic acid for 4 weeks had improved endurance and benficial changes in skeletal muscle fibre composition over those supplemented with palmitic acid [52]. Its tempting to consider the possibility of similar benefits to heart muscle. Oleic acid can reduce some problems like hepatic insulin resistance compared to a palmitic acid diet [103].

Thinking outloud: Interestingly fungus exposed to copper produces more oleic acid and less palmitate [97].

Interestingly, it is possible that copper oleate can form near ambient or plausible conditions that can then diffuse through oil [58] but its not clear how this varies among other fatty acids or biological molecules. Some works have looked at copper compounds but produced with synthetic means such as electrochemistry [24]. (Oleic acid is also the material of choice in particle formation via electric discharge [50] but again hard to get context and relevant to biological systems.) A copper compound highly diffusible in lipids may have significant impact on copper distribution. The biological implications are not clear although some literature exists on its use as a biocide along with other copper compounds [93].

It is also possible that it could also act as a simple yet unique solvent although no direct evidence is available. It may have some specific attributes as a fuel although little information currently exists. Saturated and monounsaturated fats appear to be prefered by astrocytes for B-oxidation although some confusion remains about the preference for oleic or palmitic acid [23].

The phytoestrogens are both a possible source of benefit and concern [99] [115] although most concerns appear to involve sources other than olives. For example, pinoresinol, a component of virgin olive oil, may have anti-tumor effects [62].

Its worth noting that chicken, mostly baked thighs, was a consistent dietary component. While specifics vary it does contain a substantial amount of oleic acid among other components [4] [90] [89] and similarly wht the occasional dietary component turkey [14] [114] . A more careful quantitative analysis may be helpful to determine if this content was just too low to meet all dietary needs or if some other factor in the olive oil is likely to be more important.

II.2. Liquid Sunflower Lecithin

Liquid lecithin had been used in various forms earlier. Initially gel capsules were used before switching to powdered form. While MUQED diet coding conventions were still evolving, it looks like the last capsule was given on 2021-04-02 (day 18719) on a total of 656 days. A liquid form from the same manufacuturer had also been used, between 2023-03-15(19431) and 2023-12-15(19705) , which was eventually included in this modified diet. The label is pictured in Fig. 2. A major component is oleic acid [38] (monounsaturated 18:1) although there may be more polyunsaturated fats such as 18:2 as well as saturated fats with 16 and 18 lengths. Phosphatidyl choline and related components should be common to all variants.

II.3. Inositol

Inositol literature was also highly suggestive of a wide range of benefits but no compelling evidence exists of specific efficacy and endogenous production is often considered sufficient. A 2016 report found that inositol was likely safe for dogs and cats but lacking any benefit [2]. Overall regulation of synthesis, uptake, and elimination is not well uderstood [101]. Inositol and derivatives are widely distributed in plant and animal cells and therefore exist in most diets but may be difficult to absorb while the kidneys make several grams per day in humans [64] without dietary inputs. Inositol phosphates form a family of signalling molecules [63] , including pyphosphates [104], which are common in eukaryotic cells [55] . Carefully regulated inositol derivative amounts are needed for proper development of Drosophila melanogaster [85] and an inositol hexaphosphate kinase may have an effect on lifespan in mice [72].

It is important to note that high inositol has been associated with both heart and kidney pathology suggesting it has a pathological role [80] and it " stimulates the transcription of genes underlying hypertrophic growth." [26]. However, many results show beneficial effects of exogenous inositol or phosphate derivatives.

Thinking outloud: common fallacy, see the copper paper, large amounts in disease state does not imply causing evil could be good response etc. See carnitine section comes up again with serum and cardiac divergence.

One recent review considers heart and brain disease, diabetes, cancer and reproductive health [21]. Some reports of heart or circulatory benefits include moderation of pathological effects of high fat diet on heart including improved rhythm and reduced remodeling [54]. It may also act indirectly by improving oxygen delivery from hemoglobin [13] and it does interact with RBC membranes [81]. It is thought to be useful for managing PCOS due to insulin signalling roles but clinical data are inconclusive [30]. Added myo-inositol has improved results, at least in part due to improved mitochondria performance, when added during in vitro culture of porcine embryoes [46]. Energy production was one central concern for Happy and motivation for copper supplementation. Inositol inhibits mitochondrial fission apparently due to the inositol sensing effects of AMPK [43]. Signalling by inositol phosphates is an integral part of mitochondria Ca regulation [36] [119]. Other benefits exist for conditions such as diabetic nephropathy [95] and cisplatin exposure [117] [83]. It may also have a role to play in thyroid health especially in cases of boarderline low hormone output [12] as has been suspected previously to be a common problem in dogs.

Inositol and derivatives may mitigate various stresses in a range of organisms such as salt stress in the plant Tamarix ramosissima [57] and acute ammonia toxicity in fish [112].

Inositol's interaction with lithium may be of passing relevance in most people and animals but it may illustrate some common limitations of designing interventions from incomplete pathways. It has been observed to reduce effects of toxic lithium dosing in several measures of heart quality [53]. In an unrelated development, lithium, with a long history as a drug, has recently been considered in Alzheimer's Disease due to lower amounts in the diseased state and the benefits of supplementation [9]. General effects of lithium on the brain have been explored for some time and it was suggested in 2013 and 2014 that the effects of lithium may be due to modulation of inositol signalling [107] [106]. If this is the case, the inositol system may be a better point of intervention as a problem could be fixed more directly. It is easy to find reasons why Li may be depleted in old age such as reduced stomach acid which may reduce absorption of many metals. Suppplementation may coincidentally modulate a related pathway as lithium appears to do in the case of inositol but it does not fix the actual problem (if there is an inositol defect) and therefore is likely to have limited clinical benefit. Although it may be quite useful up to a point it is also misleading and ultimately a dead end for a cure. Generally nutrient deficincies are expected [68] and inositol may be one component of a set of supplements worth investigation for this condition. This line of thought gets back to the important role of understanding cause and effect in designing an intervention but recognizing complete pathways are not known and a lot of guessing may be the most scientific thing to do.

Thinking outloud: or at least most engineering thing where the whole point is to design a solution not just get data

II.4. PABA

PABA is probably one of the more controversial components. It has been described as "vitamin like" in a report suggesting improvement in tumor radiosensitivity related to phenotype changes including reduced melanogenesis[116]. It is structurally similar to sodium benzoate which had previously been used due to a variety of speculative considerations from thyroid to dental to microbial health [69] [67] . PABA however may have coontrasting microbial and thyroid effects. It seems to have no value to the mouse but does create susceptibility to the parasite Plasmodium berghei [45] although it is also claimed to be anti-viral [7]. When combined with a tumor inducer, PABA at .5 percent of diet increased incidence of thyroid tumors and increased TSH and proliferation alone [42]. Its not clear however if these changes are pathological although suggestive of a route to cancer or other diseases. In 1948 hypothyroid conditions induced by PABA or other means showed interactions between choline deficiency and liver pathology [39]. That work is somewhat similar to the present work except that more pathways have been worked out and, "we've got computers" allowing for many possible interactions to be considered in a tractable way.

It appears to modulate neurotransmitter levels with recently described anti-cholinesterase activity [41] and enhancement of tryptophan hydroxylase activity [20]. It is used controversially in Peyronie's disease where it is thought to improve oxygenation and decrease fibrosis although side effects such as hepatitis are a concern [88].

In 1945 it was observed to improve resistance to decreased barometric pressure [37].

II.5. Choline

Choline is not considered a likely suspect due to the egg and lecithin content of the background diet but could not be ruled out. Some experiments have suggested a non-monotonic relationship to heart failure with both high and low intake being deleterious with harm mediated by a metabolite, TMAO, on the high side [79]. Similar results were shown for ASCVD [59] but association studies of this type can easily be misleading as colinearlity or antagonism with other food components could modulate apparent response to "choline" intake against this background diet. Another work showed an L shaped association between choline intake and stroke risk saturating at around 276mg/day [111]. Choline supplements may reduce the severity of Alzheimer's Disease [109]. Lack of choline in parenteral feeding appears to create liver steatosis which can be prevented with supplementation [94]. Choline is essential for dogs and commercial foods are supplemented from sources such as choline chloride [27]. Choline bitartate has recently been associated with stones in dogs [66] and so an alternative, CDP choline was used. This has been used in various experiments with dogs [118] [51] with no obvious acute issues.

II.6. Silica

Additional suspects included silicon dioxide. The biology of silicon is not well known and generally "sand" is considered inert. Silicon appears to have been well established as a nutrient with a structural or metabolic role or both. It may be beneficial in many conditions [17] [70] [16] and age related diseases such as atherosclerosis [110] although evidence is inconclusive. Very recently a silicate based therapy for dilated cardiomyopathy was investigated [102].

Candidates for supplementation include silicon dioxide and sodium metasilicate both or which are readily available. The metasilicate is fairly soluble but corrosive and concerns about safety of side reactions argued against its use. Finally, a food grade anti-caking sio2 source was purchased [1] although adding to dog diets was delayed. Silica appears in the Carlson copper supplement and possibly others. Many sources have been discussed in the literature along with suggested dosing. Interestinghly, alcohol free beer absorption was quite high and as expected absorption from colloidal silia rather low in one small test [100]. In a work exploring collagen and proteoglycan silicon, it was suggested that about 25mg of silicon/day is beneficial and orthosilicic acid as possible supplement source [87]. Another work recommended similar intake amount from grains, vegetables, and beverages made from grains[74]. Doses as high as 20mg/kg/day have been explored in postmenopausal animals and demonstrated increased bone mineral density [82]. Synthetic silicon delivery molecules such as "monomethylsilanetriol, a monomeric, organosilicon molecule [$Si(OH)_3CH_3$] have been designed around the details of the GI tract [92].

Silicon chemistry is probably best known from integrated circuit processing or silicone formulations. A common form of silicon, silicon dioxide, may differ in details of structure and purity with changes in dissolution and surface properties. In high concentrations, hydroxide and fluoride may both dissolve such sources at appreciable rates [15] but as optics people know even fingerprints can stain glass and organic acids have some dissolution abilities mostly at higher pH [11].

High doses raise concerns about silicon containing precipitates in many places. Silicate urinary stones occur in dogs although not as commonly as other types [77] [84] and can be induced in humans with supplements containing silicon sources [31].

II.7. Creatine

Creatine is quite popular, apparently safe for dogs, and may enhance a rate limiting step in energy production. One small feeding trial added .2g/kg BW to the diet of active dogs and found only a small benefit [18]. However, the more important issue is any ability to help those that seem to be impaired such as Happy. Creatine content in dog foods can vary based on composition and processing [28]. While the background diet for these dogs includes raw and processed ingredients its difficult to guess effective intake amounts.

II.8. Carnitine

Finally, carnitine HCl was also added. A small 2017 giving creatine at 250mg/day to Labrador retrievers demonstrated benefits in activity levels and body composition [108]. Infused carnitine could reduce ST elevation and improve ATP levels in experimentally ischemic dogs[32]. L-carnitine supplementation even demonstrated efficacy in DCM despite high serum concentrations as there was a cardiac deficiency [49] giving another example of how plasma concentrations can be misleading. It has been shown to go up with liver disease [73] and you have to suspect some signalling issues, as well as the possibility of simple leakage when gradients exist, driving the blood levels independently of intracellular amounts. Free carnitine was observed to increase after 8 days of starvation [86]. A small group of overweight dogs had lower plasma carnitine than normal weight peers [98].

II.9. Late Additions: Essential Fatty Acids Sources

Later into the modifications, the dog owner suggested krilloil which motivated a quick examination of fatty acid requirements for dogs. While the 2019 era diets did not include these components, the contents of commercial foods likely vary for reasons including storage conditions [48] so its possible that even if sufficient intake occured before it was marginal and not being repeated. Previously, the effects of copper on nutrient destruction were a concern [69]. The work cited above [48] also notes that oxidized lipids may be detrimental to the dogs and high linoleic acid intake can prevent conversion of ALA to EPA and DHA so freshness and balance may be important. Fatty acids considered important for dogs include linoleic acid (LA), alpha-linoleniec acid (ALA) , docosahexaneonoic acid(DHA) , arachidonic acid (AA) amd conjugated linoleic acids [10].

Thinking outloud: its worth noting ALA is also used for alpha lipoic acid which can be toxic to dogs

Some analyses of possible oil sources of fatty acids have been published [76] although as with all natural products contents will vary. Similar analyses exist for krilloil [35] and canola oil [56]. Interestinly, krill oil has been evaluated for stability with several metal sources finding that organic metal may be more stable than sulfates and the phosphatidylcholine may improve resistance to oxidation [113]. This topic is of immediate interest any be valuable later looking at limitations of diet chemistry on improving nutrition in old age.

Krilloil (2026-02-20 day 20504) and canola oil (2026-02-27 day 20511) were added intermittently. As with other classes of nutrients, intake alone was only part of the concern. Interactions, uptake, and anti-nutrient creation could matter.

Common Name	Structure	Approx. Dietary Req.
Linoleic Acid (LA)	18:2 (ω-6) [78]	2.8% to 5% of DM*
Linoleic Acid (LA)	18:2 (ω-6) [78]	3.3g/1000kcal AAFCO puppies [22]
Alpha-linolenic Acid (ALA)	18:3 (ω-3)	0.08% to 0.2% of DM
Alpha-linolenic Acid (ALA)	18:3 (ω-3)	0.2g/1000kcal AAFCO puppies [22]
Oleic Acid	18:1 (ω-9) [78]	No set minimum; usually > 10% of fat
EPA / DHA	20:5 / 22:6 (ω-3)	0.05% combined for growth/reproduction
EPA / DHA	20:5 / 22:6 (ω-3)	.1g/1000kcal AAFCO puppies [22]
Arachidonic Acid	20:4 (ω-6)	Synthesized from LA (Essential for cats, not dogs)
Total Fat		21.30g/1000kcal AAFCO puppies [22]

*DM = Dry Matter basis. Requirements vary based on life stage (Puppy vs. Adult) and activity level.

TABLE III. **Gemini generated table edited** . Essential and Functional Fatty Acids for Canine Nutrition. To relate metabolizable energy to body weight, a formula such as $.41 = ME(MJ)/BW(kg)^{(3/4)}$ [105]

	Canola Oil	**Soybean Oil**	**Olive Oil**	**Krill Oil**	**Chicken Thigh***	**Comm. Kibble****
Essential (n-6)						
Linoleic (LA)	19.0	51.0	9.7	1.5	20.0	15.2
Linoleic (LA)	21 [61]					
Linoleic low (LA)			3 [44]	3 [78]		
Linoleic high (LA)			20 [44]	50 [78]		
Essential (n-3)						
Alpha-Linolenic (ALA)	9.1	6.8	0.7	0.5	1.0	1.1
Alpha-Linolenic (ALA)	11 [61]					
EPA / DHA	0.0	0.0	0.0	25.0	0.6	0.3
EPA / DHA				30 [91]		
Monounsaturated						
Oleic	63.3	22.8	71.3	11.0	38.0	28.5
Oleic	61 [61]		80 [47]	27 [78]		
Saturated						
Palmitic	3.6	10.5	11.3	22.0	25.0	20.1
Palmitic	4 [29]		15 [44]	20 [78]		
Stearic	1.8	4.5	2.4	1.5	7.0	10.4
Stearic	2 [29]					
*Potentially Harmful****						
Erucic	0.01	0.0	0.0	0.0	0.0	0.0
Erucic high (C22:1)	3.7[40]			0.8[78]		

*Chicken thigh values represent the lipid profile of meat and skin combined.
**Commercial kibble values are estimates for standard poultry-based adult maintenance diets.
***Erucic acid is specifically relevant to the brassica family (rapeseed/canola); it is absent in animal fats and most other seed oils.

TABLE IV. **Gemini Generated** Fatty Acid Profiles of Canine Dietary Sources (% of Total Fat) with manually added literature values. Some refs [78] contain ranges and include trace components which may eventually prove to be important beneficially ot as toxins. Single values are reported here to fit format and reflect approximate average discarding outliers but all natural products are highly variable. Oil values are generally percentages of total fat contents.

III. DIET MODIFICATION DESCRIPTION

This manuscript was prepared as diet details and results were still evolving. Some text may be a bit confusing if there were mistakes during editing. A table of some diet details is shown in Table V. For full details, the data are available in various stages of processing on github, see Supplemental Information section.

Serial	Date	Event
20331 (-)	2025-08-11	First olive oil of 2025, single day
20354 (0)	2025-09-23	Start regular EVOO , day 0
20354 (0)	2025-09-23	Start regular inositol
20379 (25)	2025-10-18	Last marrow although not common before
20385 (31)	2025-10-24	Last snack until 2025-11-02
20394 (40)	2025-11-02	Resume snacks
20397 (43)	2025-11-05	Begin silica
20408 (54)	2025-11-16	First creatine
20420 (66)	2025-11-28	First liquid lecithin
20430 (76)	2025-12-08	Start carnitine (small amount prior day)
20430 (76)	2025-12-08	Last full day of snacks
20441 (87)	2025-12-19	Restart snacks

TABLE V. Dietary changes made during this work

Also note in the above table the sunflower lecithin powder was swapped to a liquid formulation more like that used in 2019. The distinction was removed from the reports initially and was changed to reflect this detail while researching this work. The label for the NOW liquid sunflower lecithin is included as Fig. 2.

Its also worth noting marrow was largely discontinued soon after Annie died on 2025-10-14. Apparently however it is also a good source of fats including oleic acid among others [6].

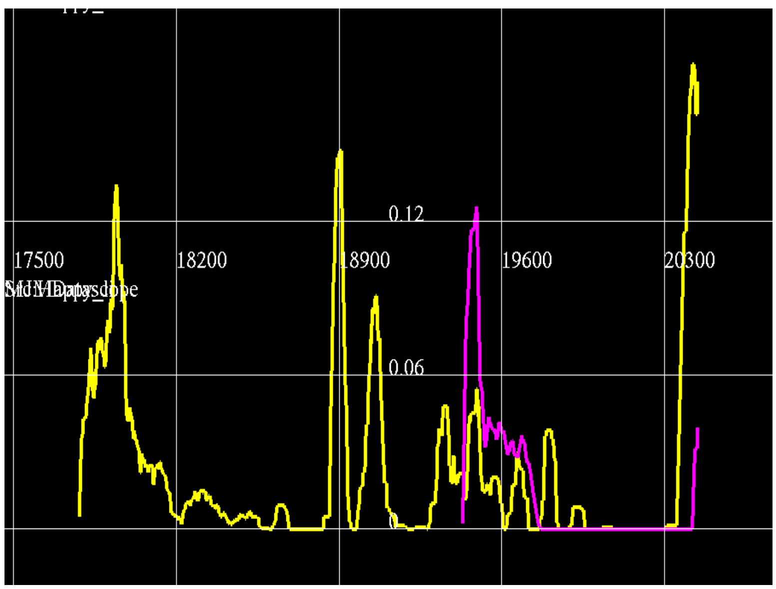

FIG. 3. Trailing 50 day average for olive oil(yellow) and 60 day for the liquid lecithin(magenta) for Happy. Past exposure and modifictions in last few months are show. Units are day number and nominal tsp. 18900=2021-09-30 , 19600=2023-08-31, 20300=2025-07-31

These of course were given in the context of a varying background diet. The monthly diet characteristics, along with some outcomes, are described in the appendix Appendix C for several contrasting months. In the Spring of 2019 Happy had a cough free period that was never well repeated. The months prior to Oct 2025 she was coughing a bit with good energy but notably acting slower than in 2019 or when Daisy was first introduced to her a few years before. A variety of caveats and footnotes are associated with the diet records. For a more complete discussion see the earlier manuscript [69] . Briefly, 2 supplemented snacks were generally given each day with consistent and rotating components. Care was taken to either serve immediately or within a few hours and stored in a refrigerator due to concerns about reactions, particulary with copper supplements. Vitamins for specific snacks were selected for many considerations including mutual interactions during preparation and after eating. Commercial kibble dinners were also given but not well recorded and other components such as table scraps and yard debris were sometimes consumed. Total calories are not reflected in the records as the absollute quantities of most food items were not recorded in consistent units.

Thinking outloud: MUQED allows coding to be very flexible and may be easy to understand or quite idiosyncratic

Its also worth noting that in December and January somtimes borax was used to wash dishes as oil accumulated on plastic wear and lecithing blobs were hard to remove. Although all of it was washed off the effects of residues can not be known.

IV. RESULTS

Adding olive oil (EVOO), liquid lecithin, and incidental silica, CDP choline, and PABA appeared to restore energy in everyone and remove minor feeding indifference in Rocky. After a couple weeks, it was noted that Happy would be quiet at time she used to cough. It will be difficult to attribute any additional results to the late additions of creatine and carnitine. In early Feb 2026 she was probably coughing more although very active compared to past. Adding lipoic acid may have helped cough and weasing.

Briefly it seemed that Rocky was congested although he spends little time exercising (mostly buried in blankets on couch or bed often with cigarette smoke) the PABA was stopped as it was the most suspicious for infection such as pneumonia. Currently he seems fine witout it although Happy may benefit from intermittent small amounts. See recent data for current dosing schedule.

Happy went through a period of reduced coughing early on the arrival of Daisy. The liquid lecithin did not have an obvious effect at the time although as is often the case the notes were not sufficient to find small outcome trends. Neither the EVOO nor inositol were considered at that time although incidental consumption could occur through foods or vitamin D which was usually mixed with EVOO.

In the month of Oct 2025 and following Rocky ate very eagerly even getting up in the morning with everyone else which he had not consistently done before although otherwsie ate vigorously. Dexter expressed more consistent eagerness although continuing to eat only one meal a day, and being somewhat limited in the amount he ate, he ate everyday which is rare for him to do over a month. He may have lost some eagerness after starting the carnitine.

The addition of creatine produced unremarkable results although it would have been difficult to observe as they all seemed to be making improvements of various kinds.

Rocky, and to a lesser extent Dexter, are not maintaining peak enthusiasm but both remain "better than before." Dexter from time to time will refuse to eat and Rocky usually eats later in the day although he prefers to eat the snack and may reject kibble now. He also seems more intersted in social aspects, wants to defend his food from the others, etc.

V. DISCUSSION

The results presented here are largely due to trial and error of things that might help based on empirical observations or suspected cause and effect that could produce beneficial outcomes. With a good empirical starting point, causality may be more useful to explore and errors more foregiving.

The initial motivation was to get Happy back to a cough free high energy state similar to 2019. While she still coughs every morning she does have more quiet periods and energy more characteristic of her before. Her foraging speed is a little slower than in 2019 but otherwise similar. It also became evident that small benefits appeared in the other dogs such as more consisent enthusiasm about eating. The limited benefit may be due to reduced lipoic acid after a new dog, Murphy, arrived in an emaciated state.

Two dogs from the prior work, Annie and Trixie, had subsequently died. Annie from kidney failure and Trixie from a large mesenteric tumor. As potential kidney and cancer benefits may exist from some of these supplements, its unfortunate they could not try this new diet and that the olive oil had not been given regularly for some time. Annie's earlier feeding hesitatancy partially resolved with zinc but its likely she had decliming kidney function for sometime as she seemed to be slowly deteriorating for much of her time here.

Thinking outloud: move to discussion or something

Trixie's last consistent exposure to olive oil ended on 2024-03-11 (19793) with only 8 additional doses before her death on 2025-09-11 (20342) . Similarly Annie died on 2025-10-14 (20375) without significant olive oil since about 2024-03-11 and that was just for a shot time. Annie had olive oil for 9 consecutive days and inositol for 6 days. There was not much apparent change and her kidney failure was thought to be disease progression and not related to the new components.

Thinking outloud: Trixie apparently had accumulated 3 liters of blood in her abdomen due to bleeding through a mesenteric tumor. Although the emergency vet said it was too coagulated to aspirate, apparently that was not the case later. The high dose vitamin K may have been working somewhat but the tissue architecture did not support resolution. Failure to continue clotting could be due to liver limitations or a consumptive coagulopathy. Asymptotic effects on coagulation at huge vitamin K doses appear to depend on specific type but intermediate plateau, limited by classical coagulation factors like proteins S an C, may not be entirely flat. Optimistically if could be regression associated bleeding as she was generally getting better based on activity interests.

It is possible that their decline was related to lack of the components considered here.

Right now it appears that Dexter both needs zinc and should avoid even minimal carnitine supplements. He avoids raw beef and lost his appetite once the carnitine was started. No specific reason for this can be identified but its possible that idiosyncracies like this exist and are even common. The raw beef mixed in with the snack does not bother him although he rejects egg white pieces.

It will be interesting to see if these new components reduce his need for supplemental zinc and fix a more fundamental problem. Ideally he will eat both snacks and get his copper again more regularly.

Thinking outloud: once formulated split into discussion as approporiate

Its been difficult to give him any copper since he only eats the riboflavin snack now but that may be worth adding back to his overall diet too as apparently the lack of zinc was a bigger problem than any probable copper toxicity.

Confusing results with inositol, a prescursor or skeleton for several signalling molecules, are probably due to adaptive responses under a variety of other limitations or states. That is, the heart is supposed to grow to accomodate additional load as occurs with exercise. Growth may occur due to cell growth, proliferation, or deposition of various matrix proteins. Each of these may be limited by different resources and have different outcomes on heart function. Inositol status may not be able to achieve a more beneficial result in all cases just reduce one which appears to be pathological. Signalling may be a best guess over evolutionary time scales for reproductive age individuals but may have some flaws.

Since inositol can be synthesized and is plentiful in the diet its important to consider why supplementation would be beneficial especially if age related and not generally provable. Age related GI atrophy has been considered before and the steps needed to free and absorb inositol investigated for potential bottlenecks. While internal synthesis could also decrease with age, its also possible, although not suggested, that requirements increase with age.

Most of the components considered had something to do with a colloquial ”energy level.” Copper, iron, carnitine, creatine, fats, iodine, etc, may be ”energy limiting” in various states. Another general feature may be ”frailty” which will be a concern more in old age diseases. There is some indication that unsaturated fats such as oleic acid can stimulate thermogenesis in brown adipose tissue [33]. Indeed, Rocky and some of the others did feel warmer on occaision although that may have been the case with more iodine in the past too.

The contributions of these components to clinical outcomes remains undertermined but empirically this appears to work well on this group of dogs. Its much easier to work backward from a known working solution and tweak values or remove things than it is to look for small signals in a trial of a partial solution. Given a group of N non-interacting components with only one being actually useful, it would be possile to find ”the one ” in $log_2(N)$ trials instead of N trials by testing half of the possible candidates at once. Mixing things always creates interaction risks but testing one partial solution at a time may not provide enough signal to be informative in cases where the components interact more than additively. In all of these cases, the hope is to find unexpected but stark error signals unless of course a good solution emerges by luck. Yet its still possible to chase a local optimum. In real life, where many interactions among candidates can be favorable, it probably is more difficult to test one at a time. While it may sound like a well controlled test, digging a small signal out of statistical noise, which is not really noise to each individual, may waste a lot of time. The signal choice and its optima in diet space need to always be reconsidered.

Happy probably needs many components to stop coughing and some keep falling through the cracks. Lipoic acid was reduced on the arrival of a malnourished dogs, Murphy, around 2026-01-03 (20456) and added back in February 2026. While retaining her energy she is back to coughing a bit.

Thinking outloud: need citations of course, case reports on other cause hypoglycemia and see if there is a pattern of symptom severity. Sugar per se is not a clinical endpoint and the dillema here illustrates a problem using blood levels of anything to inform intervention. Earlier in the snack design thoughts, lipoic acid was considered as an insulin mimetic in dogs. In one case of accidental poisoning in a dog , blood sugar reached undetectable (<1.1 mmol/L [20 mg/dL]) and remained low for days yet the dog survived[60]. Note of course that blood level does not reflect velocity and consumption could be high with production almost keeping up. In which case cells are burning a lot of it and in no need of any other fuel. Its also possible that sugar can be as much a nuisance as a fuel if cells can live with little flux and use mostly fats. While clinical hypoglycemia apparently occurs, it may be the case that lipids can largely replace sugar as fuel under the right conditions. As lipoic acid is thought to increase uptake however, high velocity consumption is likely the case.

Previously, I considered the ability of lipoic acid to reduce blood sugar to control cancer. It may be that inositol could be of benefit here with reduced blood sugar and better signalling which could allow cells to better fit into their environment. Insulin resistance is thought to precede and facilitate cancer [5] and that may be partially causal but also related to associated signalling systems. There is some reason to consider inositol as a possible factor.

V.1. Picky Eating

Thinking outloud: needs citations but could be a work of its own mostly based on personal experience although some claims can be documented

This set of ingredients made all six dogs eat enthusiastically as they were all generally more animated. As with copper, energy and "well being" may precede reversal of adaptations to nutrient imbalances such as weak or enlarged tissue. Eating "dysphoria" here has been a matter of degrees with diverse suspects. With mild amounts absent a specific cause like dental pain, considerations may include mild liver or kidney insufficiency or effective deficiency of some nutrient. Some benefits have been seen with zinc, probably more likely when ALP is near zero, and sodium benzoate which can enahance flavor, moderate microbes in mouth and stomach, and sequester amino acids. Zinc may also be suspected with a high copper diet as used here. The current set of suspects may also be broadly applicable to older dogs due common features of aging. The oleic acid may be a part of making eating more enjoyable although reasons remain unknown currently. Improved absorption of some nutrients may be one possibility.

V.2. Its the Greatest Derivative Charlie Brown

The literature on nutrition and perhaps medicine in general is filled with confusing or disappointing results. Its possible that a lot of anecdotes about folk remedies have some basis but the critical components are not always identifed leading to tests of single nutrients that may not even be given to a population in the right state. That is, having the right "background diet" or "other stuff" that makes the component under test have an observable impact on outcomes.

Dose-response curves for entities of interest are going to show a maximal benefit at some dose while becoming perhaps toxic at high doses. It is often pointed out but not fully appreciated that this "typical" curve applies to specific current states and is just a section of a larger surface. In the right state, the curve could "flat line" or even invert. The result of a randomized trial would depend on the population being explored. For any given state, there is probably only a small number of entities with easily measured positive slopes. In the case of the dogs in this work , a prior success was identified and the suspects were easy to test.

Consider the case of inositol and thyroid hormone. While inositol may have a signalling role, it can't replace say iodine or amino acids required to respond to TSH. Tested on a general population of hypothyroid patients in isolation it may have no effect yet the underlying cause of low inositol may be a combintation of other nutrional issues.

Making some guess about WHY a specific nutrient is deficient may lead to a single or simple cause that depletes many nutrients all of which need to be replaced to observe a robust recovery.

In some ways, nutrition is probably like a recurring story of Ivermectin and covid-19. That is, if you test it in a population that is widely in a state treatable with Ivermectin and that state impairs a response to covid-19, the Ivermectin empirically shows a covid benefit in that population. There is no requirement it be virus directed. Knowing the MOA resolves the issue.

VI. CONCLUSIONS

Thinking outloud: still kind of bland

The initial diet from 2019 turned out to be a remarkably good starting point but the important features were not appreciated. These results point to the importance of including extra virgin olive oil or maybe simply a source of oleic acid above some minimum amount. It was amazing though that there was a many-months time lag from initial deprivation to appearance of even minor symptoms. Solutions to problems involving nutrients likely will need to address several of them as well as interactions. The current diet appears to work well for this group of dogs and its unfortunate Annie and Trixie did not have a chance to try it. While its likely luck contributed to the early success, simply combining things seems to be a good strategy as long as incompatibilities are considered. A picky eater may have specific needs for a vitamin mix and not just be reacting to a taste preference. Its difficult to guess why the olive oil or other additions made eating more appealing but the fact that it seems to work satisfies most of the immediate objectives. Happy is still coughng but with almost youthful energy. While it is difficult to point to understanding of cause and effect when so many details remain unknown, empirical observations and hypothesized causal links may be informative. Often, these tend to be glib and simply rehashing the "usual suspects." In this case, a wider search was considered for cause and effect behind the observations. While not entirely successful it does create a route for progress.

VII. SUPPLEMENTAL INFORMATION

Dog diet data files are available online at https://github.com/mmarchywka/dogdata or other locations as may be required. The author may also be contacted if onlines sources are not avialble. Some MUQED code is available too although probably not in a state that is easy for others to build and only written for Linux, https://github.com/mmarchywka/muqed_util The source code for this paper is available at https://github.com/mmarchywka/inooil for verficiation of bibliography or other checks. Please credit or retain credits for any info or bibliographies downloaded.

VII.1. Computer Code

```
2036 ./run_linc_graph -dt-mo txt/happy3.txt
2037 texfrag -include xxxtable
2038 cp xxxtable /home/documents/latex/proj/inooil/keep/oldtablelec.tex
2039 ./run_linc_graph -dt-mo txt/happy4.txt
2040 cp xxxtable /home/documents/latex/proj/inooil/keep/newtablelec.tex
2041 history

2023 ./run_linc_graph -dt-mo txt/happylec.txt
2024 cp xxxtable /home/documents/latex/proj/inooil/keep/lectablelec.tex
```

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1. Pubmed eutils facilities and the basic research it provides.

2. Free software including Linux, R, LaTex etc.

3. Thanks everyone who contributed incidental support.

Appendix A: Statement of Conflicts

No specific funding was used in this effort and there are no relationships with others that could create a conflict of interest. I would like to develop these ideas further and have obvious bias towards making them appear successful. Barbara Cade, the dog owner, has worked in the pet food industry but this does not likely create a conflict. We have no interest in the makers of any of the products named in this work.

Appendix B: About the Authors and Facility

This work was performed at a dog rescue run by Barbara Cade and housed in rural Georgia. The author of this report ,Mike Marchywka, has a background in electrical engineering and has done extensive research using free online literature sources. I hope to find additional people interested in critically examining the results and verify that they can be reproduced effectively to treat other dogs.

Appendix C: Happy Monthly Diet Tables

Monthly diet summaries for Happy typical of most dogs except for Dexter who was not eating the second snack each day and mostly missed out on copper. Full data is available online, see Supplemental Information section. Early 2019 was noted for a brief period of no coughing with good activity levels. Very little coughing was observed around the arrival of Daisy. The months prior to October 2025 were high coughing and some decreasing activity as noted. The following months were mostly with the components discussed here as is shown in the table below.

Name	2019-04 Apr	2019-05 May	2019-06 Jun	2023-05 May	2025-03 Mar
FOOD					
KCl(tsp kcl)	0.12 ;0.25;30/30	0.22 ;0.25;31/31	0.2 ;0.25;18/18	0.077 ;0.062;25/25	0.092 ;0.062;18/18
KibbleAmJrLaPo	0.15 ;0.15;30/30	0.15 ;0.3;31/31	0.15 ;0.15;18/18	0.037 ;0.037;25/25	0.037 ;0.037;18/18
KibbleLogic	0.099 ;0.1;30/30	0.1 ;0.2;31/31	0.1 ;0.1;18/18	0.025 ;0.025;25/25	0.025 ;0.025;18/18
b10ngnc[c]	0.34 ;1;10/30	0.11 ;1;4/31			
b15ngnc[c]	0.078 ;1;2/30	0.39 ;1;12/31		0.043 ;0.25;4/25	
b20ngnc[c]	0.53 ;1;11/30	0.26 ;1;10/31	0.25 ;1;5/18	0.25 ;0.25;17/25	
b20ng			0.056 ;1;1/18		
b7ngnc[c]	0.33 ;1;9/30	0.27 ;1;9/31	0.74 ;1;13/18	0.14 ;0.25;10/25	
blueberry				2.1 ;3;14/25	
canned		0.097 ;1;3/31			
carrot	0.6 ;1;30/30	0.56 ;1;31/31	0.53 ;1;18/18	0.37 ;0.25;25/25	0.37 ;0.25;18/18
cb20ngnc					0.43 ;0.25;18/18
cbbrothbs				1.00e-02 ;0.25;1/25	
cbbroth				0.25 ;0.25;16/25	0.21 ;0.12;18/18
citrate(tsp citrate)	0.12 ;0.25;30/30	0.22 ;0.25;31/31	0.2 ;0.25;18/18	0.05 ;0.062;25/25	0.092 ;0.062;18/18
clbrothbs					0.22 ;0.12;18/18
ctbrothbs	0.72 ;1;29/30	0.79 ;1;31/31	0.6 ;0.25;17/18	0.12 ;0.25;8/25	
ctbroth				0.052 ;0.25;3/25	6.94e-03 ;0.12;1/18
eggo3	0.23 ;0.25;30/30	0.24 ;0.25;31/31	0.2 ;0.12;17/18	0.065 ;0.12;25/25	0.035 ;0.062;10/18
eggo					0.028 ;0.12;7/18
egg					3.47e-03 ;0.062;1/18
garlic	0.25 ;0.25;30/30	0.25 ;0.25;31/31	0.25 ;0.25;18/18	0.2 ;0.25;20/25	1.6 ;1;18/18
marrow					0.28 ;0.25;14/18
oliveoil(tsp)				0.03 ;0.12;9/25	
oliveoil	0.12 ;0.25;15/30	0.1 ;0.33;16/31	0.2 ;0.25;12/18		
pepper				0.085 ;0.25;6/25	
salmon	0.28 ;1;9/30		0.19 ;0.25;6/18	0.03 ;0.25;2/25	
shrimp(grams)	3 ;20;12/30	6 ;20;28/31	3 ;10;10/18	7.3 ;16;14/25	7.6 ;8.1;17/18
sorbitol(tsp)				5.94e-03 ;0.016;7/25	
spinach	0.43 ;0.25;30/30	0.46 ;0.33;31/31	0.081 ;0.25;3/18	0.28 ;0.25;20/25	0.37 ;0.25;18/18
turkey				0.35 ;0.25;24/25	
vinegar(tsp)				0.067 ;0.062;23/25	
VITAMIN					
B-1(mg)				5.64e-03 ;0.012;18/25	3.26e-03 ;0.012;9/18
B-100(count)	0.063 ;0.25;9/30	0.075 ;0.25;10/31	0.12 ;0.25;9/18		
B-12(mg)				0.05 ;0.25;7/25	0.042 ;0.25;5/18
B-2(mg)				16 ;25;17/25	8.6 ;16;18/18
B-3(mg)	8.2 ;25;10/30	4.8 ;25;7/31	6.9 ;25;5/18	7.9 ;12;22/25	13 ;24;18/18
B-6(mg)	6.5 ;25;9/30	6.3 ;25;9/31	6.9 ;25;5/18	5 ;12;12/25	1.7 ;6.2;9/18
B-multi(count)				5.00e-03 ;0.12;1/25	
Cu(mg)	2.2 ;5;29/30	1.8 ;5;31/31	0.7 ;5;17/18	0.38 ;2;17/25	8 ;6.3;18/18
D-3(iu)				58 ;300;5/25	28 ;125;4/18
Iodine(mg)[a]	0.022 ;0.25;3/30	0.012 ;0.25;2/31	0.041 ;0.25;3/18	0.12 ;0.25;21/25	0.13 ;0.78;4/18
K1(mg)				0.7 ;2.5;13/25	1.4 ;2.5;18/18
K2(mg)	1 ;3.8;9/30	0.82 ;4.9;7/31	1 ;3.8;5/18	0.4 ;1.2;12/25	0.49 ;3.1;5/18
K2MK7(mg)				0.5 ;6.2;15/25	2.08e-03 ;0.013;4/18
MgCitrate(mg)	113 ;202;19/30	76 ;202;13/31	43 ;102;8/18	96 ;100;24/25	36 ;100;12/18
MgCitrate				1.00e-02 ;0.25;1/25	
Mn(mg)		0.36 ;2;9/31			0.1 ;0.62;3/18

TABLE VI. Part 1 of 2. Events Summary for Happy from 2019-04-01 to 2025-03-31A summary of most dietary components and events for selected months between 2019-04-01and 2025-03-31. Format is average daily amount ;maximum; days given/ days in interval . Units are arbitrary except where noted. Any superscripts are defined as follows: **a)** SMVT substrate. Biotin, Pantothenate, Lipoic Acid, and Iodine known to compete..**c)** hamburger with varying fat percentages- 7,10,15,20, etc. ..

Name	2019-04 Apr	2019-05 May	2019-06 Jun	2023-05 May	2025-03 Mar
Se(mcg)				1 ;12;3/25	
Zn(mg zn)				1.6 ;5.9;10/25	0.24 ;1.5;3/18
arginine(mg)	110 ;250;15/30	87 ;325;11/31	41 ;250;4/18	98 ;350;11/25	117 ;175;8/18
biotin(mg)$^{(a)}$	1.6 ;2.5;22/30	1.6 ;3.2;22/31	1.5 ;2.5;11/18	4.8 ;5;24/25	0.97 ;2.5;13/18
folate(mg)				0.025 ;0.12;6/25	0.017 ;0.062;5/18
histidine(mg)					
histidine(tsp)					1.74e-03 ;0.0078;4/18
histidinehcl(mg)	46 ;162;10/30	108 ;325;13/31	23 ;260;2/18	8.5 ;85;3/25	
iron(mg)					1.5 ;11;4/18
isoleucine(mg)	57 ;250;10/30	64 ;250;9/31	21 ;250;2/18	72 ;200;9/25	21 ;100;4/18
lecithin(mg)				216 ;225;24/25	331 ;225;18/18
leculecithin(lecu)	518 ;300;30/30	390 ;390;23/31	219 ;300;10/18		
leucine(mg)	174 ;162;30/30	268 ;325;31/31	146 ;162;16/18	84 ;162;25/25	68 ;162;14/18
lipoicacid(mg)$^{(a)}$	5 ;20;10/30	5.5 ;30;9/31	5.3 ;20;5/18	5.5 ;12;11/25	4.2 ;12;6/18
liqlecithin(tsp)				0.05 ;0.19;25/25	
lysinehcl(mg)	295 ;325;29/30	451 ;325;31/31	242 ;325;15/18	231 ;325;25/25	135 ;162;11/18
methionine(mg)	16 ;62;9/30	35 ;125;12/31	20 ;121;3/18	60 ;62;23/25	16 ;62;8/18
pantothenate(mg)$^{(a)}$	32 ;125;10/30	35 ;125;10/31	14 ;125;2/18	14 ;39;9/25	39 ;78;17/18
phenylalanine(mg)	72 ;125;20/30	56 ;150;14/31	27 ;125;4/18	25 ;125;8/25	6.9 ;62;2/18
taurine(mg)	423 ;250;30/30	442 ;500;31/31	458 ;250;17/18	333 ;225;25/25	138 ;225;17/18
threonine(mg)	124 ;162;26/30	92 ;162;17/31	76 ;162;8/18	169 ;325;25/25	235 ;162;18/18
tryptophan(mg)	70 ;100;22/30	93 ;130;23/31	76 ;100;11/18	75 ;150;17/25	3.6 ;38;2/18
tyrosine(mg)	16 ;125;4/30	14 ;125;4/31	6.9 ;125;1/18	52 ;100;13/25	5.6 ;50;2/18
valine(mg)	79 ;206;13/30	92 ;206;16/31	78 ;212;7/18	112 ;200;14/25	189 ;200;17/18
vitamina(iu)				1260 ;4500;9/25	312 ;2250;4/18
vitaminc(mg)				5 ;31;7/25	
vitaminc(tsp)				3.13e-03 ;0.0078;18/25	1.95e-03 ;0.0039;9/18
vitamine(iu)				11 ;50;6/25	3.1 ;19;3/18
MEDICINE					
Ivermectin	0.067 ;1;2/30	0.065 ;1;2/31			
SnAg					0.056 ;1;1/18
doxycycline	0.4 ;1;12/30				
sodiumbenzoate(mg)					16 ;49;9/18

TABLE VII. Part 2 of 2. Events Summary for Happy from 2019-04-01 to 2025-03-31A summary of most dietary components and events for selected months between 2019-04-01and 2025-03-31. Format is average daily amount ;maximum; days given/ days in interval . Units are arbitrary except where noted. Any superscripts are defined as follows: **a)** SMVT substrate. Biotin, Pantothenate, Lipoic Acid, and Iodine known to compete..**c)** hamburger with varying fat percentages- 7,10,15,20, etc. ..

Name	2025-03 Mar	2025-08 Aug	2025-09 Sep	2025-10 Oct
FOOD				
KCl(tsp kcl)	0.092 ;0.062;18/18	0.085 ;0.062;31/31	0.081 ;0.12;21/21	0.077 ;0.062;24/24
KibbleAmJrLaPo	0.037 ;0.037;18/18	0.036 ;0.037;30/31	0.036 ;0.037;20/21	0.037 ;0.037;24/24
KibbleLogic	0.025 ;0.025;18/18	0.024 ;0.025;30/31	0.024 ;0.025;20/21	0.025 ;0.025;24/24
b10ngnc[c]				0.073 ;0.25;6/24
b15ngnc[c]				0.19 ;0.25;13/24
b20ngnc[c]		0.018 ;0.12;3/31	0.22 ;0.25;20/21	0.057 ;0.062;22/24
b7ngnc[c]			0.13 ;0.5;7/21	0.12 ;0.25;8/24
carrot	0.37 ;0.25;18/18	0.37 ;0.25;31/31	0.38 ;0.5;21/21	0.38 ;0.25;24/24
cb10ngnc		0.014 ;0.12;2/31		
cb20ngnc	0.43 ;0.25;18/18	0.4 ;0.25;29/31	0.22 ;0.25;12/21	
cb25ngnc		0.014 ;0.12;2/31		
cbbrothbs			1.49e-03 ;0.031;1/21	
cbbroth	0.21 ;0.12;18/18	0.21 ;0.12;31/31		0.055 ;0.12;6/24
citrate(tsp citrate)	0.092 ;0.062;18/18	0.085 ;0.062;31/31	0.081 ;0.12;21/21	0.077 ;0.062;24/24
clbrothbs	0.22 ;0.12;18/18			
ctbrothbs		0.21 ;0.12;31/31	0.22 ;0.25;21/21	0.22 ;0.12;24/24
ctbroth	6.94e-03 ;0.12;1/18	5.04e-03 ;0.12;1/31	0.22 ;0.25;21/21	0.16 ;0.12;18/24
eggo3	0.035 ;0.062;10/18	0.054 ;0.062;27/31	0.071 ;0.25;21/21	0.062 ;0.062;24/24
eggo	0.028 ;0.12;7/18	8.06e-03 ;0.062;4/31		
eggshell		0.044 ;0.12;11/31	0.024 ;0.12;4/21	0.021 ;0.12;4/24
egg	3.47e-03 ;0.062;1/18			
garlic	1.6 ;1;18/18	0.15 ;0.25;19/31	0.071 ;0.25;6/21	0.094 ;0.25;9/24
marrow	0.28 ;0.25;14/18	0.19 ;0.25;21/31	0.083 ;0.5;11/21	0.089 ;0.25;16/24
oliveoil(tsp)			0.074 ;0.38;8/21	0.18 ;0.12;24/24
oliveoil		8.06e-03 ;0.25;1/31		7.81e-03 ;0.12;2/24
salmon		0.03 ;0.25;3/31		0.051 ;0.25;4/24
salt(tsp)		1.39e-03 ;0.0039;15/31		4.88e-04 ;0.0039;3/24
shrimp(grams)	7.6 ;8.1;17/18	3.1 ;8.1;12/31	7.3 ;8.1;19/21	6.4 ;8.1;19/24
spinach	0.37 ;0.25;18/18	0.37 ;0.25;31/31	0.38 ;0.5;21/21	0.38 ;0.25;24/24
vinegar(tsp)		0.016 ;0.12;6/31	0.024 ;0.12;7/21	0.034 ;0.062;13/24
VITAMIN				
B-1(mg)	3.26e-03 ;0.012;9/18	4.55e-03 ;0.012;23/31	6.71e-03 ;0.024;21/21	5.88e-03 ;0.0059;24/24
B-12(mg)	0.042 ;0.25;5/18	0.024 ;0.12;6/31	0.048 ;0.5;5/21	0.021 ;0.12;4/24
B-2(mg)	8.6 ;16;18/18	8.1 ;8.1;31/31	9.3 ;32;21/21	7.8 ;8.1;23/24
B-3(mg)	13 ;24;18/18	11 ;12;31/31	16 ;48;21/21	12 ;12;24/24
B-6(mg)	1.7 ;6.2;9/18	1.4 ;3.1;14/31	1.9 ;12;10/21	1.7 ;3.1;13/24
Cu(mg)	8 ;6.3;18/18	3.3 ;5;24/31	2.3 ;5;18/21	3.1 ;5;23/24
D-3(iu)	28 ;125;4/18	27 ;150;6/31	21 ;150;3/21	25 ;150;4/24
Iodine(mg)[a]	0.13 ;0.78;4/18	0.26 ;3.1;5/31	0.3 ;3.1;4/21	0.065 ;0.39;4/24
K1(mg)	1.4 ;2.5;18/18	1.2 ;1.2;31/31	1.4 ;5;21/21	0.57 ;1.2;11/24
K2(mg)	0.49 ;3.1;5/18	0.06 ;0.62;3/31		0.89 ;1.9;13/24
K2MK7(mg)	2.08e-03 ;0.013;4/18	0.013 ;0.1;7/31	9.52e-03 ;0.05;4/21	6.25e-03 ;0.1;2/24
MgCitrate(mg)	36 ;100;12/18	50 ;50;31/31	57 ;200;21/21	50 ;50;24/24
Mn(mg)	0.1 ;0.62;3/18	0.048 ;0.5;3/31	0.03 ;0.62;1/21	0.052 ;0.5;3/24
PABA(mg)			0.21 ;2;5/21	0.45 ;3.9;6/24
Zn(mg zn)	0.24 ;1.5;3/18	0.24 ;1.5;5/31	0.14 ;1.5;3/21	0.24 ;1.5;4/24
arginine(mg)	117 ;175;8/18	129 ;175;16/31	133 ;175;11/21	116 ;175;12/24
arginine(tsp)		2.02e-03 ;0.062;1/31		1.30e-03 ;0.031;1/24
biotin(mg)[a]	0.97 ;2.5;13/18	0.95 ;2.5;20/31	0.54 ;2.5;11/21	0.73 ;1.2;15/24

TABLE VIII. Part 1 of 2. Events Summary for Happy from 2025-03-01 to 2025-10-24A summary of most dietary components and events for selected months between 2025-03-01and 2025-10-24. Format is average daily amount ;maximum; days given/ days in interval . Units are arbitrary except where noted. Any superscripts are defined as follows: **a)** SMVT substrate. Biotin, Pantothenate, Lipoic Acid, and Iodine known to compete..**c)** hamburger with varying fat percentages- 7,10,15,20, etc. ..

Name	2025-03 Mar	2025-08 Aug	2025-09 Sep	2025-10 Oct
cdpcholine(mg)			5.3 ;38;4/21	9.4 ;38;11/24
folate(mg)	0.017 ;0.062;5/18	0.016 ;0.062;8/31	8.93e-03 ;0.062;3/21	0.016 ;0.062;6/24
histidine(mg)				
histidine(tsp)	1.74e-03 ;0.0078;4/18			
histidinehcl(mg)		47 ;86;21/31	58 ;183;15/21	21 ;86;11/24
inositol(mg)			1.8 ;7.8;8/21	5.7 ;16;17/24
iron(mg)	1.5 ;11;4/18	0.6 ;1.3;14/31	0.57 ;5.3;6/21	0.33 ;1.3;6/24
isoleucine(mg)	21 ;100;4/18	16 ;100;5/31	14 ;100;3/21	17 ;100;4/24
lecithin(mg)	331 ;225;18/18	330 ;225;31/31	327 ;450;21/21	328 ;225;24/24
leucine(mg)	68 ;162;14/18	73 ;162;26/31	101 ;325;20/21	83 ;162;22/24
lipoicacid(mg)[(a)]	4.2 ;12;6/18	9.7 ;25;20/31	12 ;12;18/21	13 ;25;23/24
lysinehcl(mg)	135 ;162;11/18	110 ;162;15/31	120 ;325;11/21	107 ;162;12/24
methionine(mg)	16 ;62;8/18	14 ;31;14/31	16 ;125;8/21	14 ;31;11/24
pantothenate(mg)[(a)]	39 ;78;17/18	25 ;39;20/31	22 ;39;12/21	21 ;39;13/24
phenylalanine(mg)	6.9 ;62;2/18	14 ;62;7/31	27 ;250;6/21	16 ;62;6/24
taurine(mg)	138 ;225;17/18	112 ;112;31/31	107 ;112;20/21	112 ;112;24/24
threonine(mg)	235 ;162;18/18	239 ;162;31/31	244 ;325;21/21	215 ;162;24/24
tryptophan(mg)	3.6 ;38;2/18	5.8 ;28;13/31	6.5 ;55;7/21	3.4 ;14;9/24
tyrosine(mg)	5.6 ;50;2/18	7.3 ;50;8/31	6 ;25;5/21	5.2 ;25;5/24
valine(mg)	189 ;200;17/18	187 ;200;30/31	167 ;200;19/21	150 ;200;22/24
vitamina(iu)	312 ;2250;4/18	254 ;1125;7/31	161 ;1125;3/21	281 ;1125;6/24
vitaminc(tsp)	1.95e-03 ;0.0039;9/18	3.78e-04 ;0.002;6/31	8.37e-04 ;0.002;9/21	5.70e-04 ;0.002;7/24
vitamine(iu)	3.1 ;19;3/18	4.506e+04 ;1.396e+06;6/31	13 ;133;2/21	28 ;266;3/24
MEDICINE				
Drontal(mg)		1.1 ;34;1/31	1.6 ;34;1/21	
Ivermectin		0.032 ;1;1/31		0.042 ;1;1/24
SnAg	0.056 ;1;1/18			
sodiumbenzoate(mg)	16 ;49;9/18	4 ;25;5/31	5.9 ;25;5/21	9.2 ;25;9/24
RESULT				
weight(lbs)				0.73 ;18;1/24
advecta		0.032 ;1;1/31		

TABLE IX. Part 2 of 2. Events Summary for Happy from 2025-03-01 to 2025-10-24A summary of most dietary components and events for selected months between 2025-03-01and 2025-10-24. Format is average daily amount ;maximum; days given/ days in interval . Units are arbitrary except where noted. Any superscripts are defined as follows: **a)** SMVT substrate. Biotin, Pantothenate, Lipoic Acid, and Iodine known to compete..**c)** hamburger with varying fat percentages- 7,10,15,20, etc. ..

Name	2025-09 Sep	2025-10 Oct	2025-11 Nov	2025-12 Dec
FOOD				
KCl(tsp kcl)	0.081 ;0.12;21/21	0.077 ;0.062;24/24	0.086 ;0.062;29/29	0.08 ;0.062;9/9
KibbleAmJrLaPo	0.036 ;0.037;20/21	0.037 ;0.037;24/24	0.037 ;0.037;29/29	0.029 ;0.037;7/9
KibbleLogic	0.024 ;0.025;20/21	0.025 ;0.025;24/24	0.025 ;0.025;29/29	0.019 ;0.025;7/9
SiO2(mg)			1.7 ;3.1;16/29	1.4 ;3.1;4/9
b10ngnc[(c)]		0.073 ;0.25;6/24		
b15ngnc[(c)]		0.19 ;0.25;13/24	0.36 ;0.25;29/29	0.28 ;0.25;8/9
b20ngnc[(c)]	0.22 ;0.25;20/21	0.057 ;0.062;22/24	0.058 ;0.062;27/29	0.097 ;0.25;7/9
b7ngnc[(c)]	0.13 ;0.5;7/21	0.12 ;0.25;8/24	0.013 ;0.25;2/29	
carrot	0.38 ;0.5;21/21	0.38 ;0.25;24/24	0.37 ;0.25;29/29	0.33 ;0.25;9/9
cb20ngnc	0.22 ;0.25;12/21			
cbbrothbs	1.49e-03 ;0.031;1/21			
cbbroth		0.055 ;0.12;6/24		
citrate(tsp citrate)	0.081 ;0.12;21/21	0.077 ;0.062;24/24	0.086 ;0.062;29/29	0.08 ;0.062;9/9
ctbrothbs	0.22 ;0.25;21/21	0.22 ;0.12;24/24	0.22 ;0.12;29/29	0.19 ;0.12;9/9
ctbroth	0.22 ;0.25;21/21	0.16 ;0.12;18/24	0.22 ;0.12;29/29	0.19 ;0.12;9/9
eggo3	0.071 ;0.25;21/21	0.062 ;0.062;24/24	0.065 ;0.12;29/29	0.042 ;0.12;5/9
eggo				0.028 ;0.062;4/9
eggshell	0.024 ;0.12;4/21	0.021 ;0.12;4/24	8.62e-03 ;0.12;2/29	0.014 ;0.12;1/9
garlic	0.071 ;0.25;6/21	0.094 ;0.25;9/24	0.14 ;0.25;16/29	0.17 ;0.25;6/9
marrow	0.083 ;0.5;11/21	0.089 ;0.25;16/24		
oliveoil(tsp)	0.074 ;0.38;8/21	0.18 ;0.12;24/24	0.23 ;0.19;28/29	0.17 ;0.12;9/9
oliveoil		7.81e-03 ;0.12;2/24	9.70e-03 ;0.19;1/29	
salmon		0.051 ;0.25;4/24	0.044 ;0.25;5/29	
salt(tsp)		4.88e-04 ;0.0039;3/24		
shrimp(grams)	7.3 ;8.1;19/21	6.4 ;8.1;19/24	2.8 ;8.1;10/29	1.8 ;8.1;2/9
spinach	0.38 ;0.5;21/21	0.38 ;0.25;24/24	0.37 ;0.25;29/29	0.33 ;0.25;9/9
turkey				0.17 ;0.25;5/9
vinegar(tsp)	0.024 ;0.12;7/21	0.034 ;0.062;13/24		
VITAMIN				
B-1(mg)	6.71e-03 ;0.024;21/21	5.88e-03 ;0.0059;24/24	3.24e-03 ;0.012;15/29	2.61e-03 ;0.0059;4/9
B-12(mg)	0.048 ;0.5;5/21	0.021 ;0.12;4/24	0.013 ;0.12;3/29	0.028 ;0.12;2/9
B-2(mg)	9.3 ;32;21/21	7.8 ;8.1;23/24	8.4 ;16;29/29	9 ;16;9/9
B-3(mg)	16 ;48;21/21	12 ;12;24/24	12 ;24;29/29	13 ;24;9/9
B-6(mg)	1.9 ;12;10/21	1.7 ;3.1;13/24	1.6 ;6.2;14/29	1.4 ;3.1;4/9
Cu(mg)	2.3 ;5;18/21	3.1 ;5;23/24	3 ;5;28/29	1.9 ;5;5/9
D-3(iu)	21 ;150;3/21	25 ;150;4/24	31 ;150;6/29	17 ;150;1/9
Iodine(mg)[(a)]	0.3 ;3.1;4/21	0.065 ;0.39;4/24	0.08 ;0.39;6/29	0.043 ;0.39;1/9
K1(mg)	1.4 ;5;21/21	0.57 ;1.2;11/24	0.62 ;1.9;14/29	1.2 ;2.5;8/9
K2(mg)		0.89 ;1.9;13/24	0.71 ;3.8;14/29	0.14 ;1.2;1/9
K2MK7(mg)	9.52e-03 ;0.05;4/21	6.25e-03 ;0.1;2/24	3.45e-03 ;0.05;2/29	0.011 ;0.05;2/9
MgCitrate(mg)	57 ;200;21/21	50 ;50;24/24	50 ;100;28/29	56 ;100;9/9
Mn(mg)	0.03 ;0.62;1/21	0.052 ;0.5;3/24	0.034 ;0.25;4/29	0.069 ;0.38;2/9
PABA(mg)	0.21 ;2;5/21	0.45 ;3.9;6/24	0.44 ;3.9;7/29	0.43 ;3.9;1/9
PABA(tsp)			6.73e-05 ;0.002;1/29	
Zn(mg zn)	0.14 ;1.5;3/21	0.24 ;1.5;4/24	0.2 ;1.5;4/29	0.33 ;1.5;2/9
arginine(mg)	133 ;175;11/21	116 ;175;12/24	127 ;175;15/29	117 ;175;4/9
arginine(tsp)		1.30e-03 ;0.031;1/24		
biotin(mg)[(a)]	0.54 ;2.5;11/21	0.73 ;1.2;15/24	0.5 ;1.2;14/29	0.49 ;1.2;4/9
carnitinehclL(mg)				5.2 ;31;1/9

TABLE X. Part 1 of 2. Events Summary for Happy from 2025-09-01 to 2025-12-09A summary of most dietary components and events for selected months between 2025-09-01and 2025-12-09. Format is average daily amount ;maximum; days given/ days in interval . Units are arbitrary except where noted. Any superscripts are defined as follows: **a)** SMVT substrate. Biotin, Pantothenate, Lipoic Acid, and Iodine known to compete..**c)** hamburger with varying fat percentages- 7,10,15,20, etc. ..

Name	2025-09 Sep	2025-10 Oct	2025-11 Nov	2025-12 Dec
cdpcholine(mg)	5.3 ;38;4/21	9.4 ;38;11/24	15 ;38;18/29	15 ;38;6/9
creatine(mg)			118 ;175;15/29	165 ;175;7/9
folate(mg)	8.93e-03 ;0.062;3/21	0.016 ;0.062;6/24	8.62e-03 ;0.062;4/29	0.014 ;0.062;2/9
histidinehcl(mg)	58 ;183;15/21	21 ;86;11/24	25 ;86;12/29	14 ;86;2/9
inositol(mg)	1.8 ;7.8;8/21	5.7 ;16;17/24	12 ;31;22/29	1.7 ;7.8;2/9
iron(mg)	0.57 ;5.3;6/21	0.33 ;1.3;6/24	0.46 ;2.7;9/29	0.44 ;1.3;3/9
isoleucine(mg)	14 ;100;3/21	17 ;100;4/24	17 ;100;5/29	11 ;100;1/9
lecithin(mg)	327 ;450;21/21	328 ;225;24/24	310 ;675;27/29	
leucine(mg)	101 ;325;20/21	83 ;162;22/24	87 ;162;25/29	90 ;162;8/9
lipoicacid(mg)[(a)]	12 ;12;18/21	13 ;25;23/24	5.6 ;12;21/29	3.1 ;6.2;5/9
liqlecithin(tsp)			0.014 ;0.12;3/29	0.17 ;0.12;9/9
lysinehcl(mg)	120 ;325;11/21	107 ;162;12/24	104 ;162;14/29	108 ;162;5/9
methionine(mg)	16 ;125;8/21	14 ;31;11/24	12 ;62;10/29	17 ;62;4/9
pantothenate(mg)[(a)]	22 ;39;12/21	21 ;39;13/24	26 ;78;18/29	30 ;78;6/9
phenylalanine(mg)	27 ;250;6/21	16 ;62;6/24	17 ;62;8/29	21 ;62;3/9
taurine(mg)	107 ;112;20/21	112 ;112;24/24	116 ;225;29/29	125 ;225;9/9
threonine(mg)	244 ;325;21/21	215 ;162;24/24	224 ;162;29/29	217 ;162;9/9
tryptophan(mg)	6.5 ;55;7/21	3.4 ;14;9/24	7.6 ;14;16/29	7.6 ;28;4/9
tyrosine(mg)	6 ;25;5/21	5.2 ;25;5/24	8.6 ;50;8/29	2.8 ;25;1/9
valine(mg)	167 ;200;19/21	150 ;200;22/24	152 ;200;27/29	144 ;200;7/9
vitamina(iu)	161 ;1125;3/21	281 ;1125;6/24	233 ;1125;6/29	375 ;2250;2/9
vitaminc(tsp)	8.37e-04 ;0.002;9/21	5.70e-04 ;0.002;7/24	6.06e-04 ;0.002;9/29	8.68e-04 ;0.0039;3/9
vitamine(iu)	13 ;133;2/21	28 ;266;3/24	28 ;266;5/29	30 ;133;2/9
MEDICINE				
Drontal(mg)	1.6 ;34;1/21			
Ivermectin		0.042 ;1;1/24		
sodiumbenzoate(mg)	5.9 ;25;5/21	9.2 ;25;9/24	5.7 ;25;7/29	2.5 ;22;1/9
RESULT				
weight(lbs)		0.73 ;18;1/24		
advecta			0.034 ;1;1/29	

TABLE XI. Part 2 of 2. Events Summary for Happy from 2025-09-01 to 2025-12-09A summary of most dietary components and events for selected months between 2025-09-01and 2025-12-09. Format is average daily amount ;maximum; days given/ days in interval . Units are arbitrary except where noted. Any superscripts are defined as follows: **a)** SMVT substrate. Biotin, Pantothenate, Lipoic Acid, and Iodine known to compete..**c)** hamburger with varying fat percentages- 7,10,15,20, etc. ..

Appendix D: Symbols, Abbreviations and Colloquialisms

TERM definition and meaning

Appendix E: General caveats and disclaimer

This document was created in the hope it will be interesting to someone including me by providing information about some topic that may include personal experience or a literature review or description of a speculative theory or idea. There is no assurance that the content of this work will be useful for any paricular purpose.

All statements in this document were true to the best of my knowledge at the time they were made and every attempt is made to assure they are not misleading or confusing. However, information provided by others and observations that can be manipulated by unknown causes (”gaslighting”) may be misleading. Any use of this information should be preceded by validation including replication where feasible. Errors may enter into the final work at every step from conception and research to final editing.

Documents labelled ”NOTES” or ”not public” contain substantial informal or speculative content that may be terse and poorly edited or even sarcastic or profane. Documents labelled as ”public” have generally been edited to be more coherent but probably have not been reviewed or proof read.

Generally non-public documents are labelled as such to avoid confusion and embarassment and should be read with that understanding.

Appendix F: Citing this as a tech report or white paper

Note: This is mostly manually entered and not assured to be error free.

This is tech report MJM-2025-004.

Version	Date	Comments
0.01	2025-10-22	Create from empty.tex template
-	April 11, 2026	version 0.00 MJM-2025-004
1.0	20xx-xx-xx	First revision for distribution

Released versions,

build script needs to include empty releases.tex

Version	Date	URL

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author ={Mike J Marchywka } ,
type ={techreport} ,
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startdate ={2025-10-22} ,
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year ={2026} ,
sourcecode ={https://github.com/mmarchywka/inooil} ,
author1email ={marchywka@hotmail.com} ,
contact ={marchywka@hotmail.com} ,
author1id ={orcid.org/0000-0001-9237-455X} ,
pages ={ 30}
}

Supporting files. Note that some dates,sizes, and md5's will change as this is rebuilt.

This really needs to include the data analysis code but right now it is auto generated picking up things from prior build in many cases

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